Computation of the Cartesian Coordinates of Buckminsterfullerene

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Since its discovery, buckminsterfullerene (C_{60}) has been at the focus of growing interest. In the year 1985 Kroto et al. (*1*) reported the first spectroscopic evidence of a novel type of elemental carbon consisting of C_{60} molecules for which a highly symmetric structure was proposed. Five years later Krätschmer et al. (*2*) developed an efficient method to synthesize C_{60} in a single step. Modifications of this procedure suitable for student laboratory programs have been reported in *this Journal* (*3, 4*). Beaton (*5*) recently portrayed a technique for the construction of three-dimensional models of C_{60} and C_{70} with paper. The present work shows how the cartesian coordinates of buckminsterfullerene can be computed with any calculator that can take roots.

The carbon framework can be considered as a truncated icosahedron with a circumscribed sphere of radius r_{12} for the icosahedron. The positions of the vertices of the icosahedron must be computed as a provisional result. The cartesian coordinates of 10 of the 12 vertices of the icosahedron can be computed as follows.

$$X_k = \left(\frac{2r_{12}}{\sqrt{5}}\right)\cos k\delta$$
$$Y_k = \left(\frac{2r_{12}}{\sqrt{5}}\right)\sin k\delta$$
$$Z_k = (-1)^k\left(\frac{r_{12}}{\sqrt{5}}\right) \qquad (1)$$

for k = 1, 2, 3, ..., 10, where δ = 36°. The two remaining vertices are located on the z axis at $-r_{12}$ and r_{12} for k = 11 and k = 12.

From the coordinates of two vertices of the icosahedron connected by edges, the cartesian coordinates of two carbon atoms can be computed as follows:

$$x_n = X_j + \lambda(X_k - X_j)$$
$$y_n = Y_j + \lambda(Y_k - Y_j)$$
$$z_n = Z_j + \lambda(Z_k - Z_j) \qquad (2)$$

where

$$\lambda = \frac{r_5}{2r_5 + r_6}$$

where r_5 denotes the bond lengths in the pentagons, and r_6 represents the length of the bonds connecting pentagons.

David et al. (*6*) obtained from a structure determined by X-ray diffraction average values of 1.391 Å for r_6 and 1.455 Å for r_5. From r_5 and r_6 the radius r_{12} of the circumscribed sphere of the icosahedron can be computed as follows.

$$r_{12} = \zeta\sqrt{\frac{5+\sqrt{5}}{2}} \qquad (3)$$

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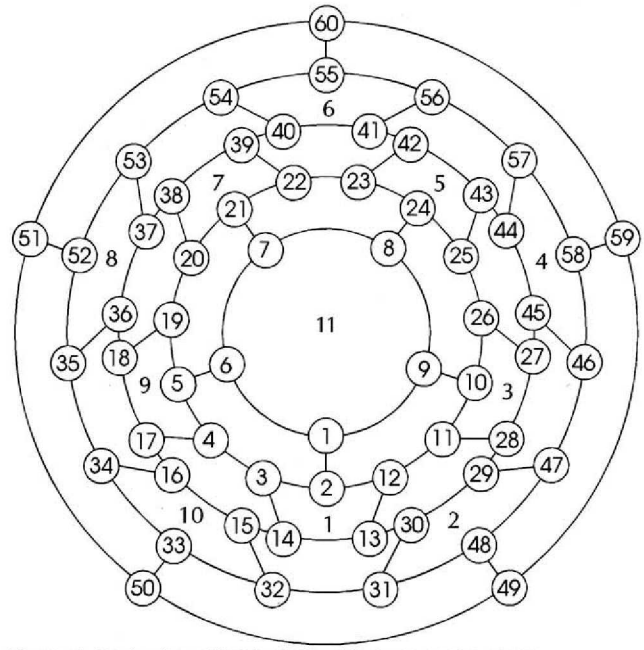

Figure 1. Projection of buckminsterfullerene on the plane.

where $\zeta = r_5 + \frac{1}{2}r_6$.

Using the experimental bond lengths reported in ref *6*, we obtain r_{12} = 4.090 Å. In buckminsterfullerene the centers of the carbon atoms are located on a sphere with radius r_B, which can be computed from r_5 and r_6 as follows:

$$r_B = \frac{1}{2}\sqrt{2(3+\sqrt{5})\zeta^2 + r_6^2} \qquad (4)$$

From the above experimental values for r_5 and r_6 we obtain r_B = 3.548 Å.

Upon truncation of the icosahedron the 20 original faces with triangular shape are converted to hexagons with alternating lengths of the sides. For each of the 12 vertices a face with the shape of a regular pentagon is formed. Because the vertices of the icosahedron have been numbered in eq 1, the same numbers can be used to label the corresponding regular pentagons. Figure 1 shows a schematic projection of buckminsterfullerene on the plane in which the carbon atoms have been placed on five concentric circles.

The pentagons are labeled with numbers as indicated above. There should be 12 pentagons labeled from 1 to 12 in Figure 1. Pentagon number 12 formed by the five atoms on the outermost circle is considered, however, to face down, and no label is shown. The 60 carbon atoms illustrated as small balls have been labeled with numbers following a spiral numbering scheme recently proposed by Taylor (*7*). In this system the carbon atoms in a fullerene are numbered consecutively from one cap of its longest

axis of symmetry through to the other cap. Figure 1 can be used to decide which vertices of the icosahedron are connected by edges and the number of the atom located with eq 2 for appropriate combinations of the indices j and k. The table below shows the combinations of j and k to be used in eq 2 to obtain the coordinates of atom number n.

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7. Taylor, R. *J. Chem. Soc. Perkin Trans. 2* **1993**, 813–824. The spiral numbering system proposed by Taylor can be applied to other fullerenes, such as C_{70}, C_{78}, etc.

The Combinations of *j* and *k* To Be Used in eq 2 To Obtain the Coordinates of Atom Number *n*.

The spiral numbering system proposed by Taylor (*7*) has been used for the carbon atoms.

n	*j*	*k*	*n*	*j*	*k*	*n*	*j*	*k*
1	11	1	**21**	7	11	**41**	6	5
2	1	11	**22**	7	5	**42**	5	6
3	1	9	**23**	5	7	**43**	5	4
4	9	1	**24**	5	11	**44**	4	5
5	9	11	**25**	5	3	**45**	4	3
6	11	9	**26**	3	5	**46**	4	2
7	11	7	**27**	3	4	**47**	2	4
8	11	5	**28**	3	2	**48**	2	12
9	11	3	**29**	2	3	**49**	12	2
10	3	11	**30**	2	1	**50**	12	10
11	3	1	**31**	2	10	**51**	12	8
12	1	3	**32**	10	2	**52**	8	12
13	1	2	**33**	10	12	**53**	8	6
14	1	10	**34**	10	8	**54**	6	8
15	10	1	**35**	8	10	**55**	6	12
16	10	9	**36**	8	9	**56**	6	4
17	9	10	**37**	8	7	**57**	4	6
18	9	8	**38**	7	8	**58**	4	12
19	9	7	**39**	7	6	**59**	12	4
20	7	9	**40**	6	7	**60**	12	6